KLE Technological University

Department of Electronics and Communication Engineering
KLE Tech, Belagavi

COURSE PROJECT REPORT

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Title:

Automated Object Removal and Image Restoration Using Stable Diffusion Inpainting

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1. Introduction & Problem Statement

The rapid growth of digital media and online content creation has significantly increased the demand for efficient image editing and restoration techniques. Unwanted objects such as people, vehicles, animals, logos, and other distracting elements often reduce the visual quality and usefulness of images. Traditional manual editing approaches are time-consuming and may produce unrealistic reconstruction results, especially when the removed region is complex or large.

There is a growing need for an automated, AI-driven solution that can intelligently identify and remove objects from images while seamlessly restoring the background. The proposed system addresses this challenge by utilizing object segmentation, mask refinement, and Stable Diffusion Inpainting to generate realistic and visually consistent outputs.

Problem Statement

> ***"To develop an AI-powered image inpainting system that automatically removes unwanted objects and generates realistic background reconstruction using Generative AI techniques."***

The system aims to simplify image editing while improving restoration quality through state-of-the-art Generative AI techniques. It provides a user-friendly web-based interface powered by Gradio, making it accessible to both technical and non-technical users.

2. Objectives

The following objectives were defined to guide the development of the AI-powered image inpainting system:

1. To develop an AI-powered image inpainting system for automated object removal from digital images.
2. To implement automatic object segmentation and mask generation using deep learning techniques.
3. To perform mask refinement using OpenCV-based morphological operations to improve mask quality before inpainting.
4. To reconstruct missing image regions using the Stable Diffusion Inpainting model for high-quality background restoration.
5. To generate realistic and visually consistent output images that match the surrounding texture, lighting, and structure.
6. To provide a user-friendly, web-based interface using Gradio for seamless image processing and real-time visualization.

3. Literature Survey

A comprehensive review of existing literature on image inpainting and object removal was conducted to understand state-of-the-art methods, their strengths, and limitations. Four key research works were studied:

3.1 Face Image Inpainting with Evolutionary Generators

This work proposed EG-GAN (Evolutionary Generator GAN), a model designed specifically for face image inpainting. The generator used an autoencoder with 256×256 face and mask input, while the discriminator evaluated full image realism using WGAN-GP. A matcher critiqued masked region consistency. The model was trained on the CelebA-HQ high-quality face dataset (24,102 training images; 2,942 test images). Performance was measured using PSNR (pixel-level reconstruction accuracy) and SSIM (structural and perceptual similarity). Higher values in both metrics indicate better inpainting quality.

DATA	METHODOLOGY	MODEL	RESULTS
CelebA-HQ dataset Image size: 256×256 Training: 24,102 images Test: 2,942 images	Generator: Autoencoder (face + mask) Discriminator: WGAN-GP realism check Matcher: Masked region consistency	EG-GAN (Evolutionary Generator GAN) Designed for face inpainting Inspired by neuro-evolution + GAN	PSNR: measures pixel-level accuracy SSIM: measures structural similarity Higher values = better quality

3.2 Object Removal and Inpainting Using Combined GANs

This paper used the CityScapes dataset (5,000 fine-annotated urban street scene images; 2,975 for training, 500 for testing) and proposed a sequential two-GAN pipeline. The first GAN removed the target object from the image, and the second GAN filled the empty space with a plausible background. Training used alternating updates with binary cross-entropy (discriminator) and cross-entropy + L1 loss (generator). Empty regions were initially filled using the Fast Marching Method (FMM) before GAN-based inpainting.

DATA	METHODOLOGY	MODEL	RESULTS
CityScapes Dataset Urban street scenes 5,000 annotated images 2,975 train / 500 test	Alternating GAN training Loss: Binary cross-entropy + L1 FMM fills empty regions initially	Object Removal GAN: removes object Inpainting GAN: fills background Sequential two-GAN pipeline	Object removal accuracy: 66.39% (real), 72.28% (fake) Inpainting accuracy: 66.41% (real), 72.63% (fake) L1 Loss: 0.7005 / 0.7032

3.3 Image Inpainting Using Deep Learning Architectures: PConv and GAN

This study used the CelebA dataset from Kaggle (202,599 facial images of 10,177 individuals, with 40 binary attribute annotations each). Two models were compared: Partial Convolutions (PConv), which focuses only on visible image pixels for more accurate gap-filling, and a GAN-based model. Preprocessing involved random masking and data augmentation. Loss functions included pixel-wise L1 loss, perceptual loss, and style loss. The GAN model outperformed PConv with SSIM of 0.84 vs. 0.82, while PConv achieved a PSNR of 15.75 dB.

DATA	METHODOLOGY	MODEL	RESULTS
CelebA from Kaggle 202,599 images 10,177 individuals 40 binary attributes per image	Random masking + augmentation Loss: Pixel-wise L1, Perceptual, Style	PConv: focuses on visible pixels only GAN: fills via adversarial competition	PConv: SSIM = 0.82, PSNR = 15.75 dB GAN: SSIM = 0.84 (better)

3.4 Image Inpainting with Structured / Checkerboard-like Masking

This paper explored structured masking (chessboard-like patterns where every alternate patch is removed) on CelebA and ImageNet datasets. The proposed method combined PGN (Progressive Growing Network) for step-by-step image filling from rough to detailed, LSTM for remembering nearby areas to maintain consistency, and GAN for realism checking. Results showed PSNR of 25.56 and SSIM of 0.82 on ImageNet, and PSNR of 26.47 with SSIM of 0.87 on CelebA.

DATA	METHODOLOGY	MODEL	RESULTS
CelebA and ImageNet	Chessboard-like masking Image resizing + normalization	PGN: progressive refinement LSTM: spatial memory GAN: realism checking	ImageNet: PSNR 25.56, SSIM 0.82 CelebA: PSNR 26.47, SSIM 0.87

4. Different Methods for Image Inpainting

Several approaches exist for solving the image inpainting and object removal problem. The following methods were evaluated before selecting the most appropriate one for this project:

4.1 Patch-Based Methods

Patch-based methods work by copying similar patches (small image blocks) from nearby regions of the image and placing them into the missing area. The system searches for the most suitable matching texture and uses it for reconstruction. While effective for repetitive textures, these methods often fail when the missing region contains complex semantic content such as faces or objects.

4.2 Deep Learning Methods (CNN-Based)

CNN-based image inpainting uses deep neural networks to learn image patterns, structures, and textures from large datasets. The model predicts the missing region based on the surrounding visual information. Partial Convolutions (PConv) are a notable CNN variant that only processes valid (unmasked) pixels, resulting in more accurate boundary reconstruction.

4.3 GAN-Based Methods

Generative Adversarial Networks (GANs) consist of two competing models: a Generator that creates the missing image region and a Discriminator that evaluates whether the generated content looks real or fake. Both models train iteratively, progressively improving reconstruction quality. GANs produce sharper results but are prone to training instability and mode collapse if not carefully balanced.

4.4 Diffusion Models

Diffusion models are advanced generative models that reconstruct missing image regions by gradually removing noise and generating new pixels based on the surrounding image context. They produce highly realistic results, are more stable than GANs, and benefit from large-scale pre-training on diverse image datasets.

5. Chosen Method: Pre-trained Stable Diffusion Model

After evaluating all available methods, the pre-trained Stable Diffusion Inpainting model was selected as the core algorithm for this project. The model used is runwayml/stable-diffusion-inpainting — an open-source model specifically trained for image inpainting tasks and available via Hugging Face. The key reasons for this choice are:

5.1 Better Quality Results

Diffusion models generate highly realistic and natural-looking output images. They understand the surrounding image context and reconstruct the missing region by maintaining proper texture, color, lighting, and structure — significantly outperforming traditional patch-based and GAN-based approaches.

5.2 No Need to Train from Scratch

The Stable Diffusion Inpainting model is already pre-trained on massive image datasets. This eliminates the need to collect large training datasets or spend computational resources on model training, making the system faster to implement and deploy.

5.3 More Stable than GANs

Training GANs requires careful balance between the generator and discriminator. If disrupted, output quality deteriorates or training fails entirely. Diffusion models are inherently more stable during the generation process, producing consistent high-quality outputs.

5.4 Easy Implementation

Pre-trained diffusion models are readily available on platforms like Hugging Face and can be directly integrated into Python applications using the diffusers library. This significantly reduces implementation complexity and development time compared to training GAN models from scratch.

6. System Architecture

The system is composed of eight interconnected modules that form a complete pipeline from image input to the final restored output. The architecture is illustrated below:

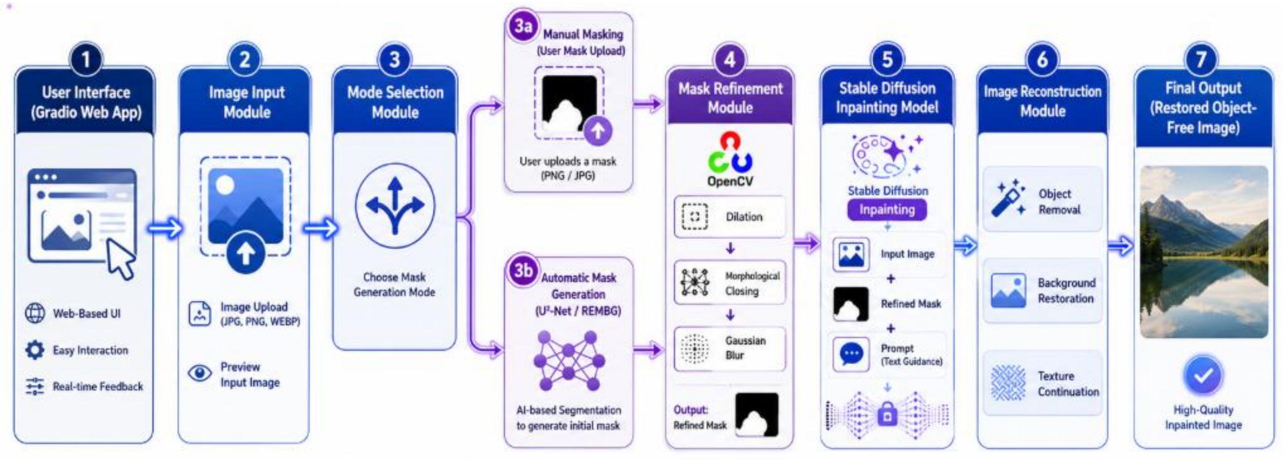

Fig 6.1 – System Architecture: End-to-End Object Removal and Image Restoration Pipeline

6.1 Image Input Module

Acts as the entry point of the system. Accepts user-uploaded images in multiple formats including JPG, PNG, and WEBP. Provides a preview of the uploaded image before processing begins.

6.2 Mode Selection Module

Provides two distinct methods for object removal, offering flexibility for different use cases:

- Manual Mask Upload: The user manually uploads a binary mask (PNG/JPG) indicating the region to be removed (white region = object to remove).
- Automatic Object Detection & Mask Creation: AI-based segmentation (U²-Net / REMBG) automatically detects the foreground object and generates the initial mask.

6.3 U²-Net (REMBG) Segmentation Model

U²-Net is a deep learning-based image segmentation model designed to accurately identify and separate foreground objects from the background. The input image is passed through U²-Net, which extracts object boundaries and foreground regions. The foreground object is then removed and a binary mask is generated. The REMBG library provides easy access to U²-Net and simplifies the segmentation implementation as a free, open-source Python library.

6.4 Mask Refinement Module (OpenCV)

The initial binary mask may contain noise, small holes, and incomplete boundaries. The Mask Refinement Module uses OpenCV morphological operations to improve mask quality:

- Dilation: Expands the mask boundaries to ensure complete coverage of the target object.
- Morphological Closing: Fills small gaps and holes within the mask region.

- Gaussian Blur: Smoothens the mask edges to remove noise and produce a cleaner boundary.

The output of this module is a refined mask ready for inpainting.

6.5 Stable Diffusion Inpainting Model

This is the core module that performs the actual object removal and background reconstruction. It accepts the original image, refined binary mask, a positive text prompt (describing the desired background), and a negative text prompt (to suppress artifacts). The working process is:

1. The masked region is identified from the refined binary mask.
2. The model analyzes nearby colors, textures, lighting, and context.
3. New pixels are generated for the masked area using the diffusion process.
4. The generated content is seamlessly blended with the surrounding pixels.

6.6 Image Reconstruction Module

Generates the final restored image by combining the inpainted region with the original image. Key operations include reconstructing the removed region, naturally continuing surrounding textures across the boundary, and matching colors and lighting between the inpainted region and the original.

6.7 Gradio Web Interface

Gradio provides a user-friendly, browser-based interface. Key features include easy image upload, mode selection between manual and automatic masking, real-time visualization of the refined mask and generated output, and lightweight deployment suitable for both local and cloud environments.

7. Web Interface

The web-based interface was built using Gradio and provides two operational modes accessible through a toggle at the top of the interface.

7.1 Manual Mask Inpainting Mode

In Manual Mode, the user uploads both the original image and a corresponding binary mask image. The white region in the mask indicates the object to be removed. The system refines the mask using OpenCV and then passes both the image and refined mask to the Stable Diffusion inpainting pipeline to generate the final output.

7.2 Automatic AI Masking Mode

In Automatic Mode, the user only needs to upload the original image. The system automatically detects the foreground object using REMBG (U^2-Net), generates a mask, refines it using OpenCV, and then passes both to the Stable Diffusion inpainting pipeline. This mode is ideal for users who do not have access to separate masking tools. The interface displays the auto-generated mask alongside the final restored image for transparency.

8. Results

8.1 Manual Mode Output

In the Manual Mode test, an image of a person standing on a desert road was used. The user provided a manually drawn binary mask highlighting the person as the object to be removed. The system successfully removed the person from the image and reconstructed the road and desert background in a visually seamless manner — maintaining road texture, mountain background, and sky continuity.

8.2 Automatic Mode Output

In the Automatic Mode test, an image of a tiger in a natural habitat was used. The REMBG model (U²-Net) automatically detected the tiger as the foreground object and generated a binary mask. After mask refinement and Stable Diffusion inpainting, the tiger was successfully removed and the background — a lush green forest with grass and trees — was realistically reconstructed without any visible artifacts or boundary seams.

Both test cases demonstrate that the system effectively removes objects and generates natural-looking backgrounds that match the surrounding scene in terms of texture, color, and lighting. The outputs confirm the viability of using Stable Diffusion Inpainting for real-world image restoration tasks.

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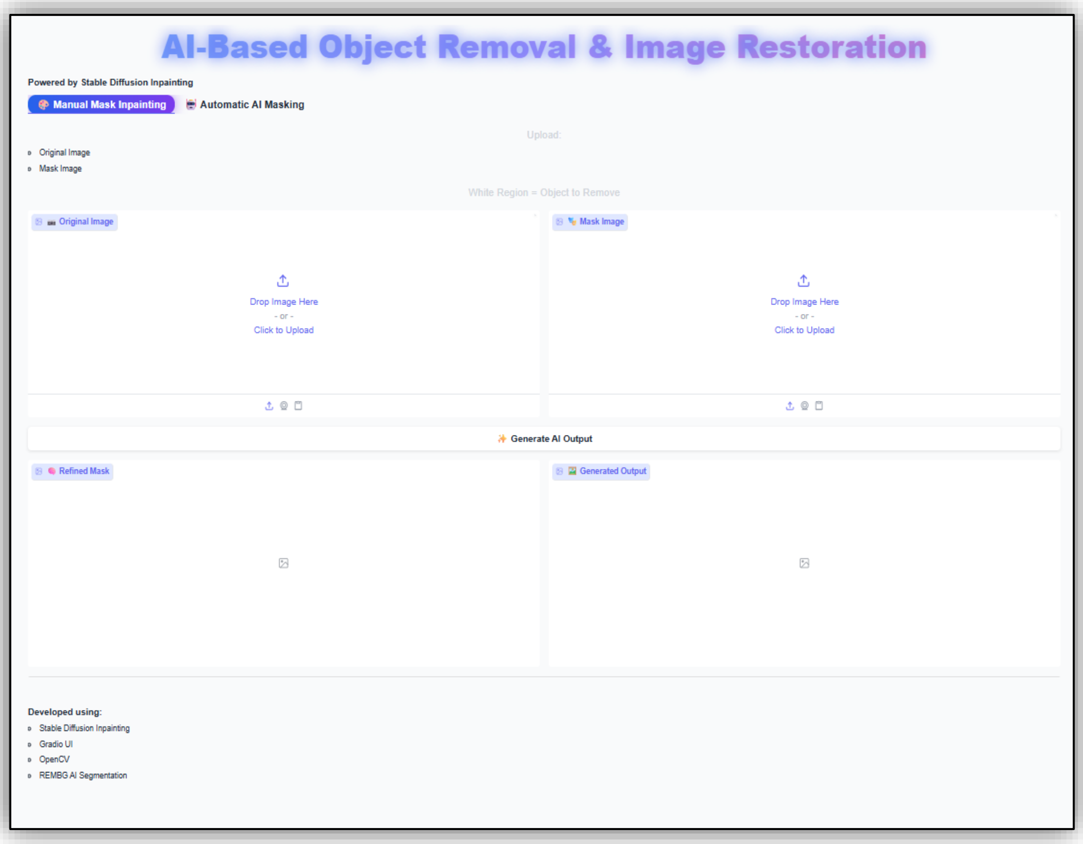
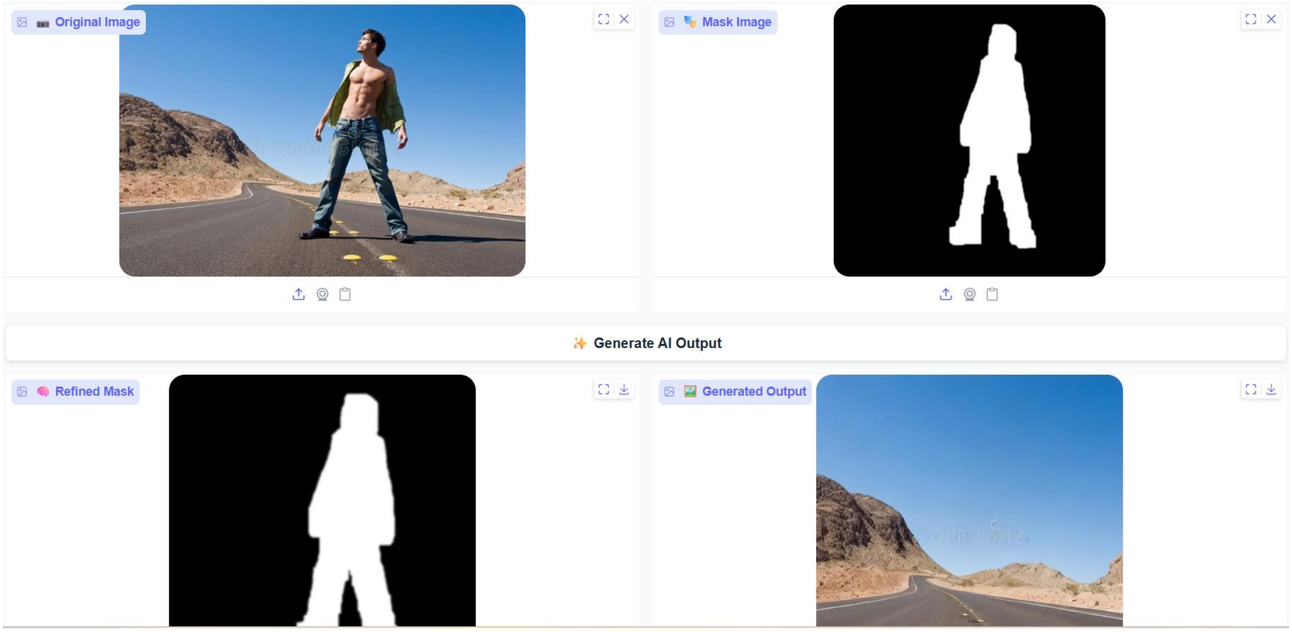
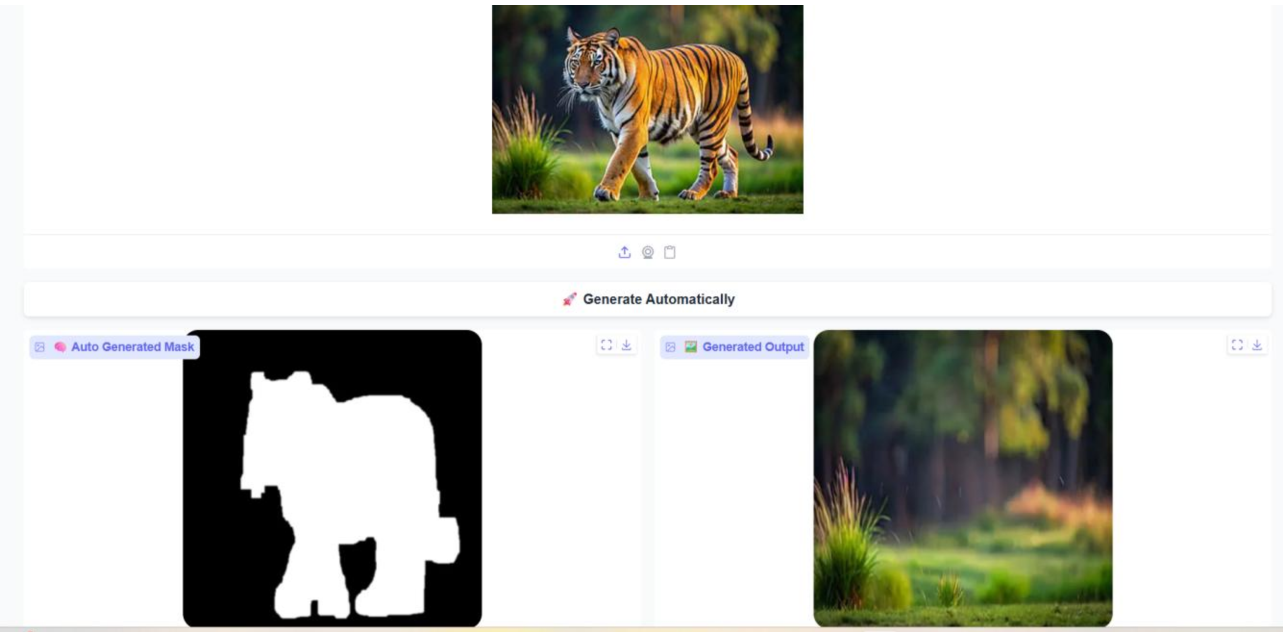
Web page interface



Manual Mode



Auto Mode



9. Model Validation

The performance of the inpainting system was evaluated using the following approaches:

9.1 Visual Comparison

Visual comparison was performed between the original image, masked image, and the final reconstructed image. The following aspects were verified:

- The missing or removed region is completely and properly filled.
- The generated output matches the surrounding background naturally without visible seams or boundaries.
- No visible gaps, distortions, or unnatural patches are present in the output image.

9.2 Quality Checking

The quality of inpainted images was assessed against the following criteria:

- The removed object is completely invisible in the final output.
- The background looks natural and contextually appropriate relative to the scene.

Texture and structural consistency are maintained at the inpainting boundary

10. Conclusion

This project successfully developed an AI-powered image inpainting system for automated object removal and image restoration using Stable Diffusion Inpainting. The system integrates multiple state-of-the-art components: U²-Net for automatic foreground object segmentation, REMBG for easy mask generation, OpenCV for mask refinement, and the runwayml/stable-diffusion-inpainting model for generating high-quality background reconstructions.

Key achievements of the project include:

- Successful implementation of both manual and automatic masking modes, providing flexibility to different types of users.
- Reliable foreground object detection across diverse image categories including humans and animals using the REMBG (U²-Net) library.
- Effective mask quality improvement through the OpenCV morphological pipeline (dilation, closing, Gaussian blur), leading to cleaner inpainting boundaries.
- Visually compelling and realistic background reconstructions using Stable Diffusion Inpainting, outperforming traditional patch-based and GAN-based methods in output quality and ease of deployment.
- A seamless, user-friendly Gradio web interface that enables non-technical users to perform object removal without any coding knowledge.

The system was validated through visual comparison and quality checks, confirming that output images are natural-looking with no visible object remnants. The results from both manual and automatic modes demonstrate the effectiveness of combining U²-Net segmentation with Stable Diffusion Inpainting for real-world image restoration.

Future work could include support for multiple simultaneous object removal, extension to video inpainting, integration of quantitative metrics (PSNR/SSIM) for automated benchmarking, and fine-tuning the diffusion model on domain-specific datasets for improved results in specialized scenarios.

— End of Report —